

EDUCATION

Master of Science in Computer Science, *SUNY - Stony Brook University*

Expected Graduation - August 2023

Bachelor of Engineering in Information Technology, *Osmania University*; GPA : 3.9/4.0

August 2015 — May 2019

SKILLS

Programming Languages	Python, Java, C++, Javascript, HTML/CSS, SQL, MongoDB
Tools & Libraries	Spring Boot, Django, Flask, React, Node.js, Bootstrap, Angular, Tensorflow, Pytorch, Keras
Cloud Technologies	Docker, Kubernetes, AWS, Azure, GCP
Relevant Courses	Data Structures, Algorithms, Operating Systems, Computer Networks, Data Science, Database Systems, Human Computer Interaction, Distributed Systems

TECHNICAL EXPERIENCE

Software Engineer - NLP

October 2020 — May 2021

BeamX TechLabs Pvt. Ltd.

Hyderabad, IN

- Designed and developed the pipeline of virtual assistant chat bot capable of engaging in dialogues and question-answering from textual/tabular documents using RASA and BERT-QA and improved the accuracy from 65% to 95% for classifying long documents into multiple classes using hyperparameter tuning.
- Implemented a RESTful API in Java to enable cookie-free Single Sign On authentication for millions of users using Keycloak IAM.
- Deployed STUN/TURN based WebRTC server as a micro-service to enable concurrent real-time audio-video communication between thousands of users.
- Dockerized all micro-service modules to deploy on Kubernetes for a 99% system uptime.
- Enhanced the API response time by 25% through remodeling Database Schema.
- Spearheaded a team of 4 peers to develop an in-house Chatbot UI kit for integration with Android.

Associate Business Technology Analyst

June 2019 — October 2020

Deloitte Touché Tohmatsu Ltd.

Hyderabad, IN

- Developed and deployed Data collection and pre-processing pipelines as AWS Lambda step functions to pre-process the Loan Notice Document Images to improve extraction accuracy.
- Overhauled the OCR back-end using Tesseract OCR which enhanced the overall extraction efficiency from 87% to 94%.
- Formulated OCR extraction patterns using regular expressions to extract data from Loan Notice Documents.
- Developed a service to automatically perform a set of system tests on extraction patterns bringing down the time to fix issues by 50%.

PUBLICATIONS

- A Novel Approach to Detect Malaria from Cell Images using Deep Learning Approaches. [View Publication](#) September 2020
- Face Emotion Recognition From Feature Extraction on NN Classifier. November 2018

PROJECTS

Fetal Heart Rate Analysis

December 2021 — Present

- Working with Prof. I.V. Ramakrishnan at Knowledge Systems Lab to model a scalable database of FHR signals and clinical data.
- Performed data analysis of large medical datasets of fetal heart rate signals to detect any neurological disorders in child caused due to events during delivery or ante/intrapartum period of mother.
- Developing a frontend query builder tool for easy extraction of data by medical practitioners without writing SQL queries.
- Tech Stack: Python, JavaScript, HTML, CSS, Django, PostgreSQL, Elasticsearch

Predicting Circadian Cycles based on Twitter Data

October 2021 — December 2021

- Performed a detailed Data Analysis of the Twitter Tweet Posts to predict the sleep pattern of the people based on their profession, weather effects and geographical locations. [Detailed Report](#)

Natural Language Query to SQL Query generation using Content Enhanced BERT

October 2021 — December 2021

- Transformed Natural Language Queries from WikiSQL Dataset into SQL queries by integrating database design principles as a vector to the BERT model. SQLNet with bi-directional LSTM was used as a baseline, and it was improved using content enhanced bert. [View Project](#)

Social Micro Blogging Web Application

September 2020

- Developed prototype of Twitter website enabling users to post tweets, follow users and in-app messaging.
- Designed 3-tier distributed web application using message queues as middleware and scaled horizontally.
- Added JWT Authentication for login using Java Spring Boot.
- Technologies: React.js, Node.js, MongoDB, MySQL, AWS EC2, AWS S3

LEADERSHIP ACTIVITIES & ACHIEVEMENTS

- Leading Computer Science Graduate Student Organisation of 500 students as a Senator.
- Among 10 awardees to get Best New Hire recognition at Deloitte FinTech Division.
- Outstanding Project Award from the Department of IT among 50 proposals for Undergraduate Final Year Major Project.
- Secured 1st Prize in IEEE South India Regional Conference Hackathon and Project Expo.